11 Double Integrals

Definition 11.1: Imprecise Definition of Double Integral

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the xy-plane. The **double integral** is defined by

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$$

where $\mathrm{d}A$ is the area element in the xy-plane.

This integral can be interpreted as the volume under the surface $z = f(x, y)$ and above the region $\mathcal{D}$ in the xy-plane.

Below are some properties of double integrals that will be useful in evaluating them.

- $\iint_{\mathcal{D}} f(x, y) + g(x, y)\,\mathrm{d}A = \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A + \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$
- $\iint_{\mathcal{D}} c \cdot f(x, y)\,\mathrm{d}A = c \cdot \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$
- If $f(x, y) \leq g(x, y)$ for all $(x, y) \in \mathcal{D}$, then $\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A \leq \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$.

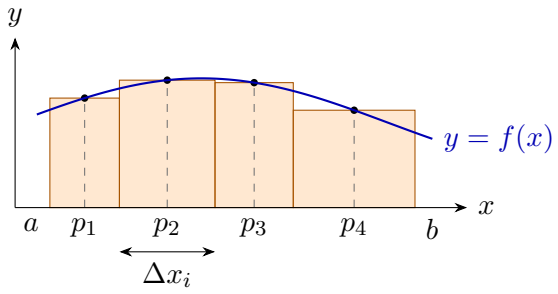
11.1 Riemann Sums and the Definition of Double Integrals

Recall the concept of Riemann sums for single-variable functions.

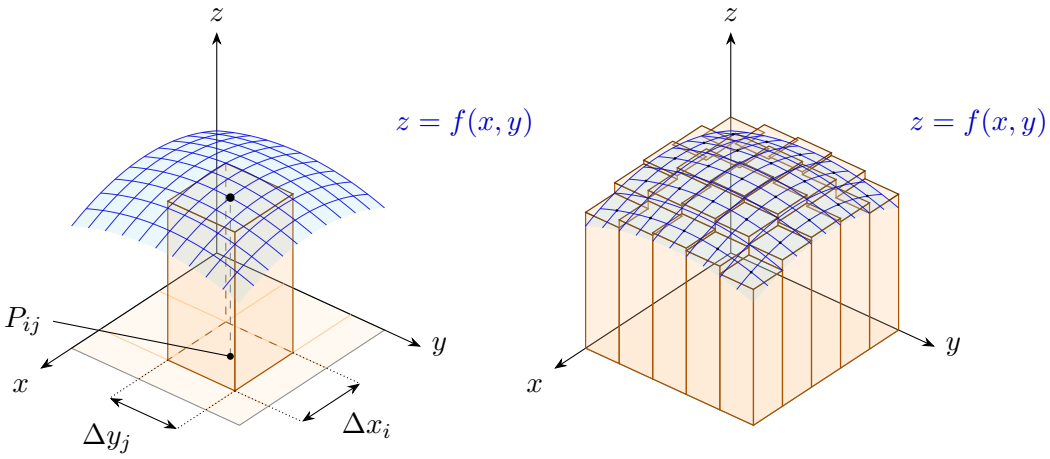
The Riemann sum for a function $f(x)$ over an interval $[a, b]$ is given by

$$\sum_{i=1}^{n} f(p_i)\Delta x_i$$

where Δx_i is the width of the i-th subinterval, and p_i is a sample point in that subinterval, as indicated in the figure below.



We can extend this idea to functions of two variables. Assume that $\mathcal{D} = [a, b] \times [c, d]$ is rectangular region in the xy-plane.



By this, we could write a precise definition of the double integral as follows:

Definition 11.2: Precise Definition of Double Integral

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the xy-plane. We partition $\mathcal{D}$ into m subintervals in the x-direction and n subintervals in the y-direction, forming a grid of rectangles.

Let Δx_i and Δy_j be the widths of the subintervals in the x and y directions, respectively, and let P_{ij} be a sample point in the rectangle formed by the i-th subinterval in the x-direction and the j-th subinterval in the y-direction.

The **double integral** of $f(x, y)$ over the region $\mathcal{D}$ is defined as the limit of the Riemann

sum as the maximum width of the subintervals approaches zero:

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \lim_{\|P\|\to 0} \sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}$$

Notice that the ΔA_{ij} here does not always exist. If it does, we call the function $f(x,y)$ **integrable** over the region $\mathcal{D}$. Also, if a function is continuous over a closed and bounded region, then it is integrable over that region.

11.2 Iterated Integrals and Fubini's Theorem

To evaluate double integrals, we can use Fubini's Theorem, which allows us to compute the double integral as an iterated integral.

Theorem 11.1: Fubini's Theorem

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$